High-Value Quality Control and Registration Validation Across 930 nm and 1100 nm Two-Photon Image Stacks

Rationale

Accurate spatial registration between 930 nm and 1100 nm two-photon image volumes is essential for reliable structural–functional comparisons, vascular–neuronal coupling analyses, and multi-fluorophore colocalization studies. Even sub-pixel misalignment can introduce systematic bias in measurements of vessel diameter, RBC velocity, bouton localization, or signal correlation across channels.

To ensure spatial concordance between wavelength-specific acquisitions, we implemented a multi-tiered quality control (QC) framework combining:

1. Intensity-based image QC metrics
2. Overlay visualization
3. Pixel-wise correlation analysis

Together, these analyses provide quantitative and visual validation of co-registration fidelity across excitation wavelengths.

```
# Path to your ScanImage stack (replace 'text' with your directory
path = '/Users/jcoleman/Documents/--LARGE DATA--/#VonRoemeling/ZT20 TIFs'

# Name of the TIF file to import
# tif_filename1 = 'ZT20_25x_930_t0_area1_00001.tif' #'ZT20_25x_930_t5_area1_00001.tif'
tif_filename1 = 'ZT20_25x_930_t0_area1_00001.tif'
tif_filename2 = 'ZT20_25x_1100_t0_area1_00001.tif' #'ZT20_25x_1100_t5_area1_00001.tif'
# tif_filename2 = 'ZT20_25x_1100_t5_area1_00001.tif'
```

High-Value QC Metrics

1. Per-Z Intensity Stability Metrics

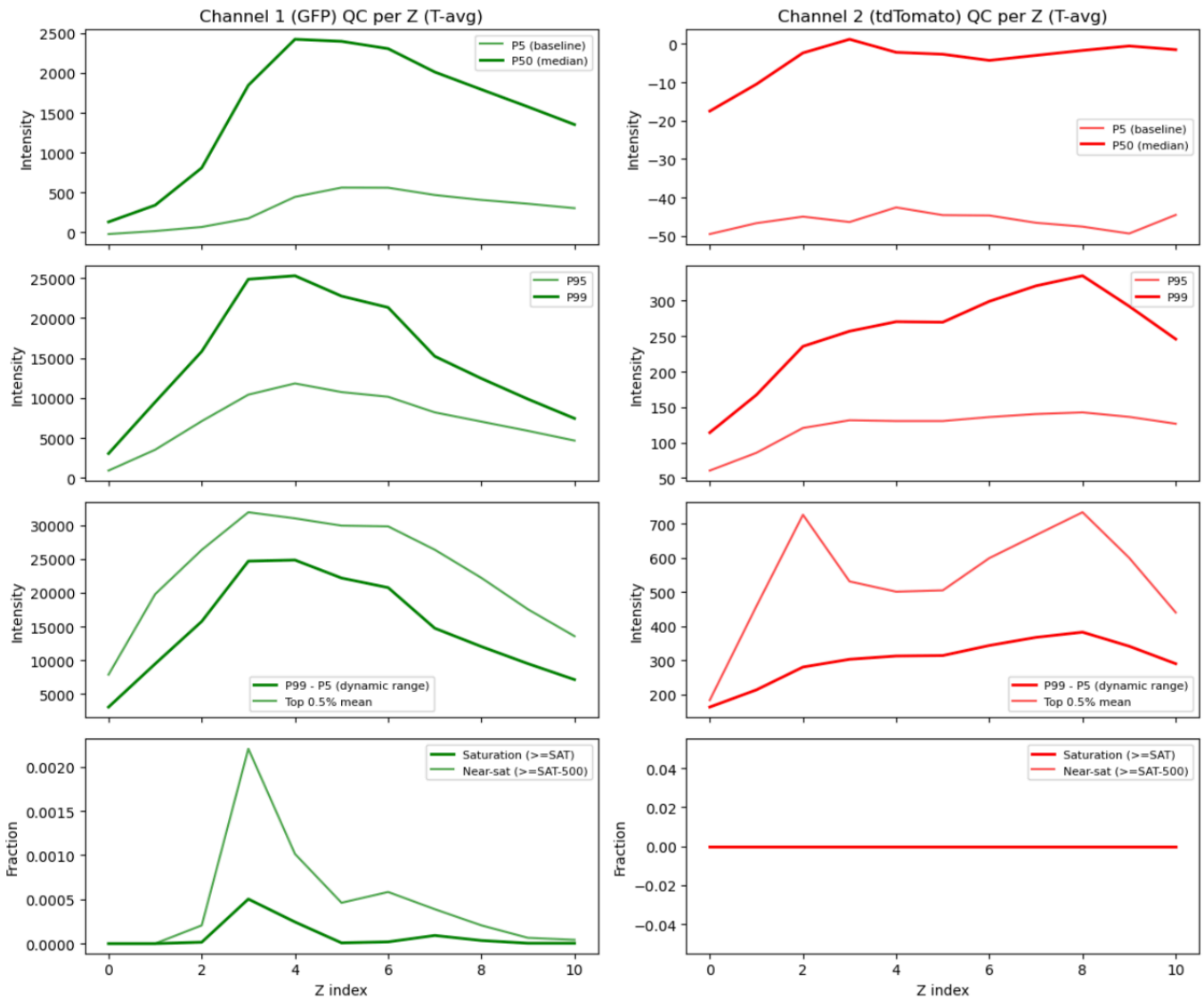
For each Z-plane (after temporal averaging across frames), we computed:

- Median intensity
- Interquartile range (IQR)
- 5th, 50th, 95th, and 99th percentiles
- Near-saturation fraction
- Top-intensity mean (upper 0.5%)

Purpose:
These metrics detect:

- Laser power drift
- PMT offset instability
- Saturation artifacts
- Axial attenuation differences
- Optical misalignment between sessions

Stable percentile profiles across Z indicate consistent optical throughput and detector behavior, reducing the likelihood that registration artifacts arise from intensity distortions.



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2. Robust RGB Overlay Visualization

For each corresponding Z-slice, we generated a dual-channel RGB overlay:

- 930 nm channel mapped to green
- 1100 nm channel mapped to red

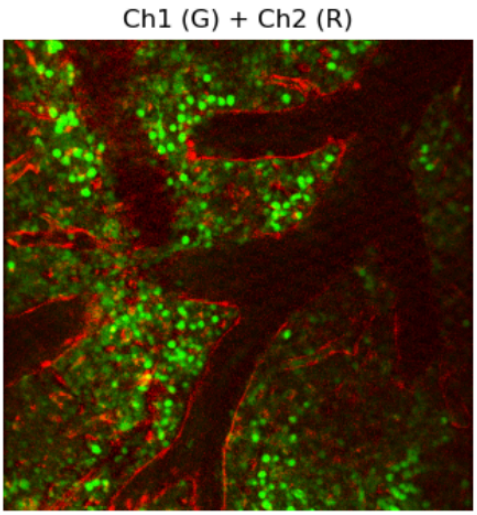
- Each channel robustly scaled using percentile-based normalization (1st–99th percentile)

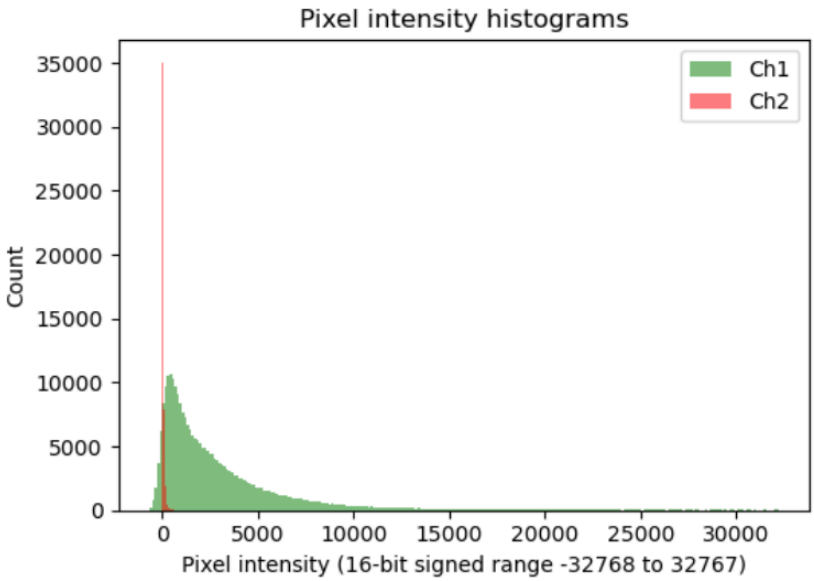
Rationale:
Percentile scaling prevents outlier-driven compression of dynamic range and preserves biologically relevant contrast.

Interpretation:

- Yellow structures → high spatial overlap
- Green-only / red-only edges → potential axial or lateral misalignment
- Color fringing → sub-pixel offset
- Spatially coherent overlap → successful registration

This qualitative check provides immediate confirmation of structural congruence.





Quantitative Registration Validation
3. Pixel-wise Intensity Correlation

For each Z-plane, we computed Pearson's correlation coefficient (r) between flattened pixel intensities:

$$
r = \frac{\text{cov}(I_{930}, I_{1100})}{\sigma_{930} \sigma_{1100}}
$$

Where:

- I_{930} = intensity vector from 930 nm image
- I_{1100} = intensity vector from matched 1100 nm image

Only corresponding spatial coordinates were compared.

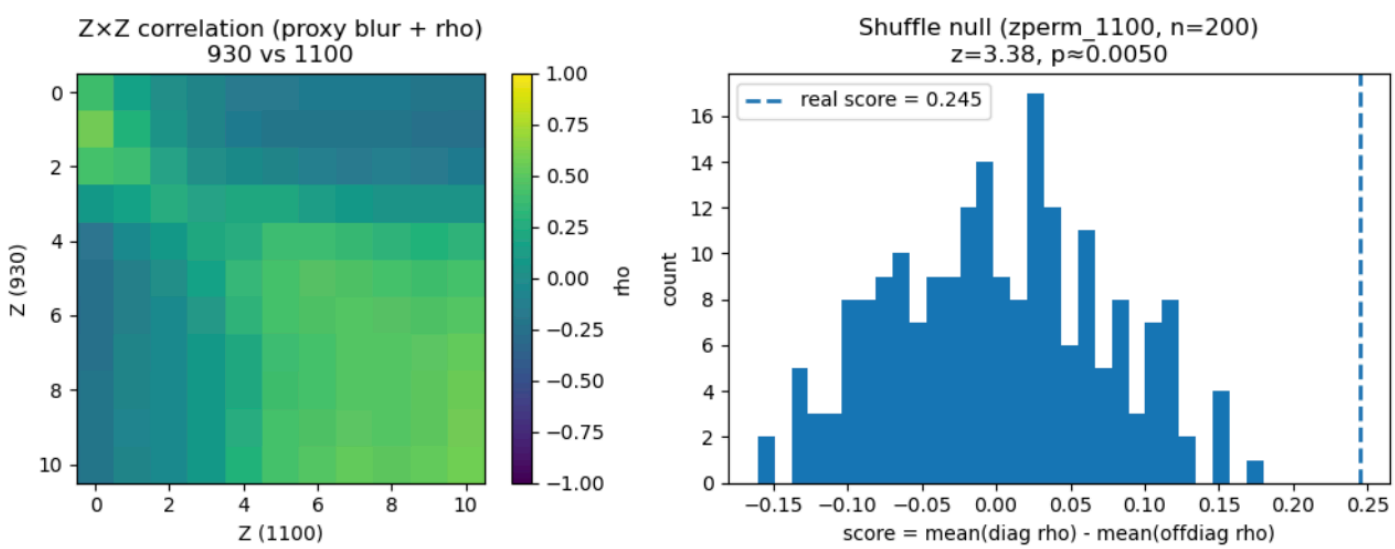
Interpretation of Correlation Values

r value	Interpretation
> 0.9	Excellent spatial registration
0.75–0.9	Strong structural concordance
0.5–0.75	Moderate overlap (possible axial shift or biological differences)
< 0.5	Misregistration or channel-specific signal divergence

High correlation across Z confirms:

- Minimal lateral shift
- No rotational misalignment
- Stable scaling between sessions
- Consistent anatomical structure alignment

Importantly, correlation is sensitive to sub-pixel shifts; even a 1–2 pixel offset produces measurable reduction in r.




```
real_score > 0.2 → PASS
real_score < 0.1 → FAIL
RESULTS:
real_score | zscore | p_empirical = (0.2446386069059372, 3.3818065788902856, 0.004975124378109453)
```

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4. Intensity Scatter Density Plots

We additionally visualized pixel-wise intensity scatter plots:

- X-axis: 930 nm intensity

- Y-axis: 1100 nm intensity
- Each point: one pixel

A tight linear distribution indicates:

- Co-registered structural features
- Linear intensity relationship
- No spatial distortion

Diffuse or multi-modal distributions may indicate:

- Channel-specific labeling differences
- Axial shift
- Optical aberration
- Nonlinear gain differences

This combined framework is powerful because it:

- Detects sub-pixel misalignment
- Identifies wavelength-dependent optical distortion
- Separates biological divergence from mechanical misregistration
- Provides both qualitative and quantitative validation
- Scales automatically across full Z-stacks

Importantly, these analyses are performed directly on raw or minimally processed data, avoiding interpolation artifacts that could artificially inflate correlation values.

Conclusion

The convergence of:

- Stable percentile intensity profiles
- Coherent RGB overlays
- High pixel-wise Pearson correlation
- Linear intensity scatter relationships

provides strong evidence for **accurate spatial registration between 930 nm and 1100 nm image volumes**.

This validation step ensures that downstream measurements — including vessel diameter, RBC flux estimation, bouton colocalization, and cross-channel structural analyses — are not confounded by acquisition-dependent spatial misalignment.